

3 All living organisms are made of cells.

Eukaryotic cells and prokaryotic cells have some features in common and some differences.

(a) (i) Which cells have a cell wall?

(1)

- ☐ **A** plant cells only
- ☐ **B** animal cells and prokaryotic cells
- ☐ **C** plant cells and prokaryotic cells
- ☐ **D** animal cells, plant cells and prokaryotic cells

(ii) Eukaryotic cells contain a nucleus.

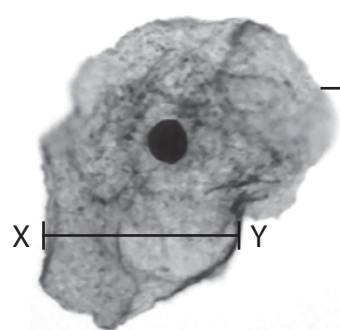
Draw a labelled diagram of a nucleus, as seen using an electron microscope.

(4)



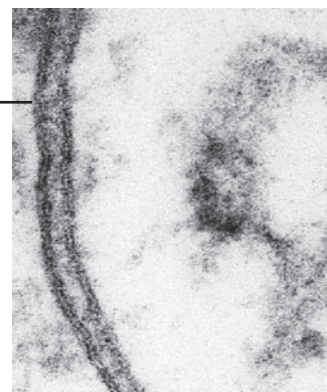
- (b) The photographs show a cell as seen using a light microscope and part of two cells using an electron microscope.

The bilayer structure of the cell surface membrane can be seen only with an electron microscope.



(Source: © Komsan Loonprom/Shutterstock)

Cell seen using a light microscope



(Source: © Dennis Kunkel Microscopy /Science Photo Library)

Part of cells seen using an electron microscope

- (i) Explain why the bilayer structure of the cell surface membrane could not be seen using a light microscope.

(2)

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- (ii) The light microscope magnified the cell 200 times.
Calculate the actual width of this cell by measuring the line XY.

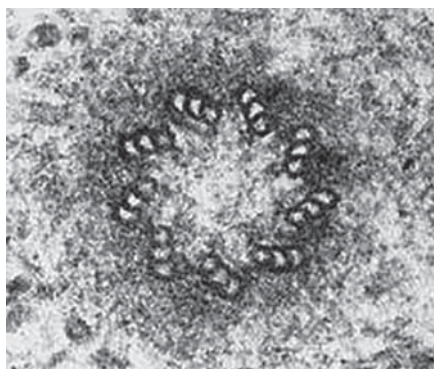
Give your answer in μm .

(3)

Answer μm



- (iii) The photograph shows an organelle that can be found in a eukaryotic cell, as seen using an electron microscope.



Describe the function of this organelle.

(2)

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(Total for Question 3 = 12 marks)

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DO NOT WRITE IN THIS AREA

